

Intelligent Medical Report Analysis System

Project Domain: DL / GenAI / NLP
DSA1 LAB Group 2



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Our *team*

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Dual-Stream Architecture

- **Extraction Pipeline:** ClinicalBERT (medicalai/ClinicalBERT) for **Token-Level Named Entity Recognition (NER)**
- **Generative Pipeline:** Fine-tuned LLM for **patient-friendly dashboard insights**

ClinicalBERT Architecture

- **Model:** BERT-base-uncased (ClinicalBERT)
- **12 Transformer Layers**
- **768 Hidden Dimensions**
- **12 Attention Heads**
- **110M Parameters**
- **Task:** `AutoModelForTokenClassification`

NER Label Space (9 Classes)

- **B/I-TEST** – Test Name
- **B/I-VALUE** – Lab Value
- **B/I-UNIT** – Measurement Unit
- **B/I-REFERENCE** – Reference Range
- **O** – Non-Entity

Training Configuration

- **IOB tagging scheme**
- **Maximum Sequence Length: 128 tokens**
- **Dynamic Padding & Truncation** using `DataCollatorForTokenClassification`

Architecture Justification – Why ClinicalBERT?

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1.Domain-Specific Knowledge:

Pretrained on **MIMIC-III clinical notes**, understands medical terminology, abbreviations, and lab values.

2.High Token-Level Precision:

AutoModelForTokenClassification extracts **Test Name, Value, Unit, and Reference Range** accurately with minimal hallucination risk.

3.Efficient & Lightweight:

110M parameters enable faster training, lower VRAM usage, and efficient deployment.

4.Robust Entity Recognition:

Uses **IOB tagging** and **subword alignment (-100 masking)** for accurate multi-token entity extraction.

Architecture	Advantage	Disadvantage	Our Assessment
ClinicalBERT	Domain-pretrained, efficient	May miss long-range context	Selected
BioBERT	Biomedical pretraining	Larger, slower	Overkill for NER
PubMedBERT	Recent PubMed pretraining	Limited clinical notes	Not MIMIC-trained
GPT/LLM	Flexible, generative	Hallucination risk, expensive	Not suitable for extraction
CRF + Word Vectors	Lightweight	Manual feature engineering	Less accurate

Baseline Model Performance

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Extraction Baseline (Heuristic vs ClinicalBERT)

Metric	Score
Accuracy	85.17%
Precision	98.64%
Recall	85.17%
Weighted F1-Score	91.27%
Entity F1-Score	24.73%

Key Observation

- Very high **Precision (98.64%)** due to dictionary matching.
- Poor **Entity F1 (24.73%)** as heuristic methods fail on multi-token entities and complex clinical text.
- Motivates the use of **ClinicalBERT** for robust NER.

Candidate Dataset for Hyperparameter Tuning

Split	Samples
Train	1,759 (70%)
Validation	376 (15%)
Test	377 (15%)
Total	2,512

Why this dataset?

- Preserves **entity density** and **label distribution**
- Maintains **document length** and **clinical note diversity**
- Enables **faster hyperparameter tuning** while remaining representative of the full dataset

Hyperparameter Tuning

Optimized Training Configuration

Hyperparameter	Selected Value
Learning Rate	2×10^{-5}
Batch Size	16
Epochs	10
Weight Decay	0.01
Warmup Ratio	0.1
LR Scheduler	Linear
Best Model Metric	Validation F1
Random Seed	42

Hyperparameters Tested

- **Learning Rate:** 1e-5, 2e-5, 5e-5
- **Batch Size:** 8, 16, 32
- **Epochs:** 3, 5, 10, 12, 30
- **Weight Decay:** 0.001, 0.01
- **Warmup Ratio:** 0.05, 0.1, 0.2

Training Highlights 6

- Training Loss: **1.237** → **0.0073**
- Lowest Validation Loss: **0.085 (Epoch 4)**
- Final Accuracy: **99.83%**
- Final Precision: **99.86%**
- Final Recall: **99.83%**
- Final F1-Score: **99.84%**

Key Outcomes

- Rapid and stable convergence within **10 epochs**
- **2e-5** learning rate achieved the best balance between convergence and stability
- Warmup + Linear LR scheduling improved training stability
- Best model selected automatically using **Validation F1**
- **Entity F1 on unseen test set: 80.53%**

BioMistral-7B: Model Selection for Fine-Tuning

Readability Baseline (7 Candidates)



Key Observation

- BioMistral-7B achieved the lowest FKGL (4.41) of all 7 models tested, with room below the Grade-8 target.
- It was also the only model to hold 100% numeric fidelity in both zero-shot and few-shot conditions.
- Motivates LoRA/QLoRA fine-tuning on BioMistral-7B to close the remaining structural-consistency gap.

BioMistral-7B Selection Metrics

Metric	Best Value
Flesch-Kincaid Grade (readability)	4.41 (zero-shot)
ROUGE-L (structural match)	0.535 (few-shot)
BLEU (phrasing match)	0.329 (few-shot)
Numeric Fidelity	100% (both conditions)

Why BioMistral-7B?

- Strongest readability and only model with zero numeric-fidelity failures
- Fits within a free Kaggle T4 GPU's memory for LoRA/QLoRA training
- Domain-pretrained (biomedical corpus), preserving clinical reasoning
- Prompting alone still traded off readability vs. structure — fine-tuning closes that gap

End-to-End Pipeline

Clinical Reports
(PDF / Scanned Images / EHR)



OCR & Text Extraction
(PyPDF2 / pdfplumber / OCR)



Text Preprocessing
• Normalization
• ClinicalBERT Tokenization
• Label Alignment



ClinicalBERT NER
Token Classification
(TEST, VALUE, UNIT, REFERENCE)



Entity Decoding
→ Structured JSON



Fine-Tuned LLM
Patient-Friendly Insights



Dashboard
• Lab Results
• Visualizations
• Health Summary

Performance Summary

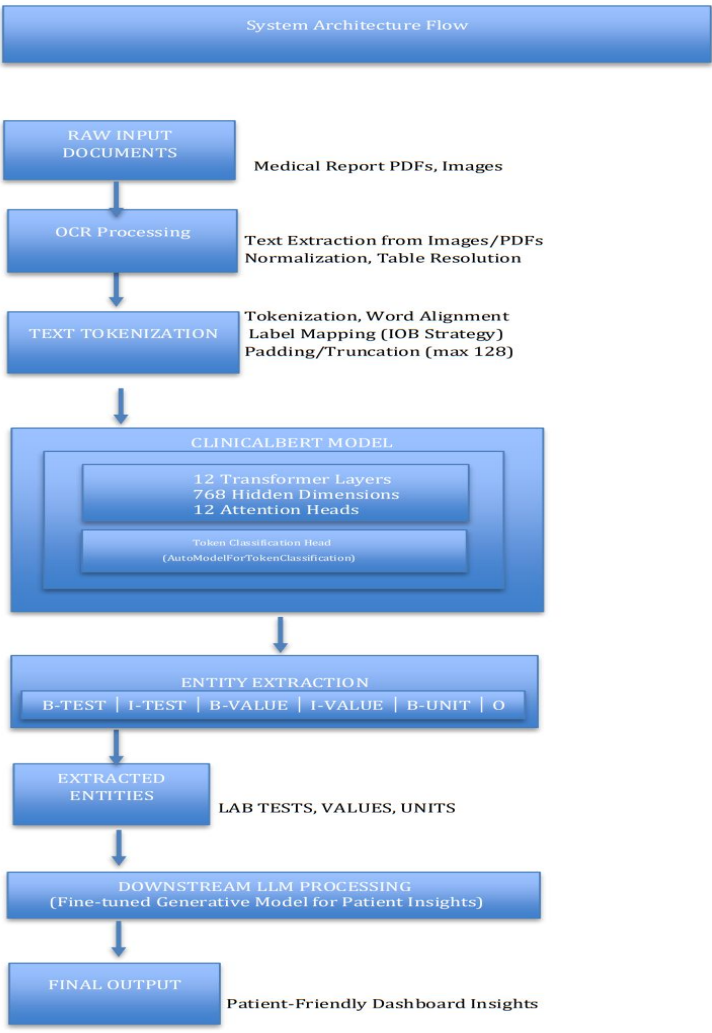
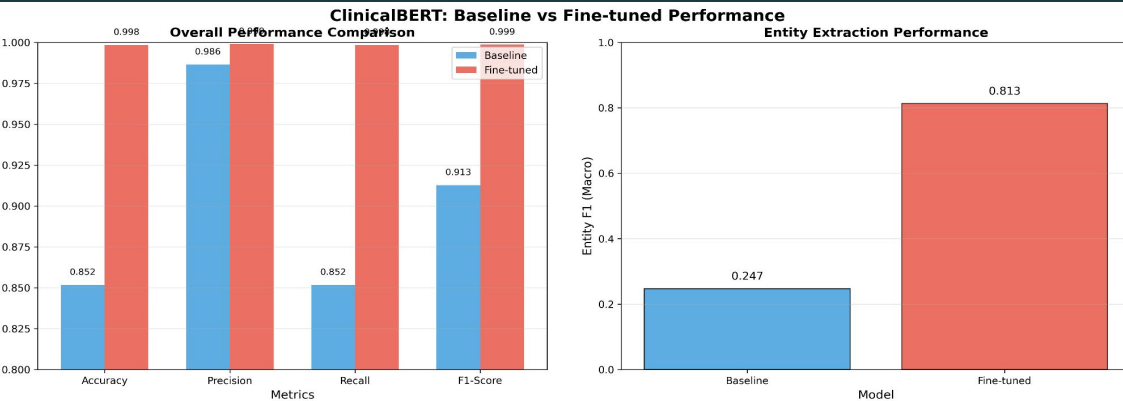
Metric	Target	Achieved
Token Accuracy	>95%	99.81%
Token F1-Score	>95%	99.84%
Entity F1-Score	>70%	80.53%
Precision	>95%	99.89%
Recall	>95%	99.81%

Key Achievement

- End-to-end pipeline successfully converts **raw clinical reports** → **structured lab data** → **patient-friendly insights** → **interactive dashboard** with **99%+ token-level accuracy**.

Results Summary

Metric	Heuristic Baseline	Fine-tuned ClinicalBERT	Improvement
Accuracy	85.17%	99.81%	+14.64%
Precision	98.64%	99.89%	+1.25%
Recall	85.17%	99.81%	+14.64%
F1-Score	91.27%	99.84%	+8.57%
Entity F1	24.73%	80.53%	55.80%



Thank you

